Intelligent Fleet Allocation System (Phase 1 & Phase 2)

Predictive Demand Forecasting, Dynamic Pricing, and Dispatch Simulation

A project report submitted in partial fulfilment of the requirements for the mid-term evaluation for the award of the degree of

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Certificate

This is to certify that the project titled “Intelligent Fleet Allocation System (Phase 1 & Phase 2): Predictive Demand Forecasting, Dynamic Pricing, and Dispatch Simulation” is a true record of the work in progress carried out by:

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under my supervision and guidance. This project meets the requirements for the mid-term evaluation for the Bachelor of Technology in Computer Science and Engineering at the Netaji Subhas University of Technology, New Delhi 110078.

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We express our sincere gratitude to Dr. Arvind Kumar for his continuous guidance, constructive feedback, and encouragement throughout the project. His insights helped us extend our earlier work (Phase 1) into Phase 2, where we introduced forecasting, pricing intelligence, and simulation-based evaluation for fleet operations.

We also thank the Department of Computer Science & Engineering, NSUT, for providing the required resources and computing environment to execute machine learning training pipelines and model experimentation. Finally, we acknowledge the support of our peers and families for their motivation and assistance during implementation and documentation.

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Abstract

Urban ride-hailing systems often experience a supply and demand imbalance due to rapidly changing passenger needs that depend on location. In Phase 1 of our project, we tackled this issue by finding spatial hotspots using unsupervised learning on historical taxi GPS data. However, simply identifying hotspots does not help with proactive fleet planning. We can improve operational efficiency when systems can predict demand, adjust pricing, and assess dispatch strategies without expensive real-world testing.

This report discusses the second phase of the Intelligent Fleet Allocation System, which adds three predictive modules:

(i) Demand Forecasting that uses an LSTM to predict demand one hour ahead for each cluster,

(ii) Dynamic Pricing employing gradient-boosted regression (XGBoost / GradientBoostingRegressor fallback) to estimate trip duration as a proxy for fare and simulate surge pricing, and

(iii) a Discrete-Event Dispatch Simulation to compare AI-driven hotspot positioning with a random-walk baseline based on wait-time metrics.

The Phase 2 system combines these modules into a unified Streamlit dashboard, incorporating the existing components from Phase 1. Preliminary results show that predictive forecasting and simulated dispatch strategies can significantly lower expected wait times during high-demand situations. This supports our goal of shifting from reactive fleet management to predictive decision-making operations.

1. Introduction

1.1 Background

Urban mobility platforms, like ride-hailing, taxis, and deliveries, need to constantly reposition their fleets to match changing demand. Demand is not evenly spread out; it tends to gather in "hotspots" shaped by land use, such as business districts, transport hubs, and campuses, as well as time factors, including rush hours and late-night peaks. A reactive dispatch policy that serves requests only after they come in often leads to longer passenger wait times and makes drivers cruise without passengers between trips. A data-driven fleet allocation system can improve service quality by:

(i) reliably identifying demand clusters,

(ii) predicting near-future demand by region, and

(iii)Assessing dispatch policies before they are put into action using simulation.

1.2 Motivation

The practical goal of this BTP is to turn large historical GPS trip datasets into a useful decision-support tool. By using spatial clustering, time-series forecasting, and simulation together, the system aids proactive fleet rebalancing decisions and offers clear dashboards for monitoring. The project also shows a complete engineering process that includes data pipelines, model training and storage, and user interface integration.

1.3 Objectives

- Phase 1 (Completed – Semester 7): Discover and validate demand hotspots using spatial clustering; compare K-Means and HDBSCAN; launch an interactive dashboard.
- Phase 2 (In Progress – Semester 8): Add demand forecasting, dynamic pricing intelligence, and dispatch simulation; integrate these modules into the dashboard; measure operational gains.

1.4 Scope of the Report (Mid-Term)

This mid-term report follows the committee's guidelines and contains: an introduction, literature review, problem statement, summary of work completed so far (Phase 1 recap and Phase 2 progress), future work plan, and references. Where final experimental plots and tables are not available, clear placeholders and measurement plans are included.

2. Literature Survey

2.1 Evolution of Urban Mobility Analytics

With more GPS traces and trip records available, researchers study mobility using spatiotemporal patterns. Important tasks include finding demand hotspots, predicting travel time, forecasting demand, and optimizing dispatch. Early efforts relied on descriptive heatmaps and rule-based zones. Today, modern systems use clustering to discover regions and machine learning for forecasting and pricing, backed by simulations for testing policies.2.2 Spatial Clustering for Hotspot Discovery

2.2.1 K-Means Clustering

K-Means divides data into K clusters by reducing the sum of squared distances within each cluster. It is efficient and allows for quick inference once the centroids are determined. This makes it a good choice for assigning new data points to clusters in real time. However, K-Means assumes that clusters are roughly spherical and needs a predetermined K, which might not represent the actual shape of city hotspots.

2.2.2 Density-Based Clustering (DBSCAN / HDBSCAN)

Density-based methods create clusters by looking at local point density and can identify sparse areas as noise. DBSCAN depends on a set distance threshold (eps). HDBSCAN expands on this by creating a hierarchy of density-based clusters and producing a stable flat clustering. HDBSCAN is effective when cluster shapes are irregular and when the dataset includes outliers or areas with varying density, which is often the case with GPS pickup data.

2.3 Taxi Demand Forecasting

Demand forecasting predicts the number of pickup requests in a region over a specific period, such as 15 to 60 minutes. Statistical methods include naive predictors, moving averages, and ARIMA/SARIMA, which both account for seasonality and trends. Deep learning models like LSTM and GRU capture non-linear relationships and long-term time

effects. A practical engineering method is to forecast demand counts at the cluster level, where clusters are formed using spatial clustering.

2.4 Dynamic Pricing and Incentive Modeling

Ride-hailing platforms often change pricing through surge multipliers. This encourages drivers to move toward areas with high demand and reduces extra demand. Pricing estimation usually relies on tabular supervised learning with features like time, location, distance, and traffic indicators. Gradient-boosting models, such as XGBoost, serve as solid starting points for tabular predictions. They also offer clarity through feature importance.

2.5 Dispatch Optimization and Simulation

Dispatch optimization assigns vehicles to requests and may move vehicles between requests. Since deployment testing costs a lot, simulation is used to evaluate policies. Discrete-event simulation models require the arrival times, vehicle availability, travel times, and service durations. Performance is measured using average and median wait time, service rate, vehicle use, and dead mileage.

3. Problem Statement

Traditional reactive dispatching and static hotspot mapping create inefficiencies in real-world fleet operations. Fleets face: (i) longer passenger wait times during local demand spikes, (ii) low driver utilization because of idle cruising and dead mileage, and (iii) a lack of ability to assess whether a demand-aware policy actually performs better than baselines before deployment. The project tackles these issues by combining strong hotspot discovery (Phase 1) with predictive demand forecasting, dynamic pricing estimation, and a simulation-based evaluation framework (Phase 2).

3.1 Research Questions

- How do K-Means and HDBSCAN compare for GPS pickup hotspot discovery regarding cluster validity and usability?
- Can we reliably forecast cluster-wise demand using sequence models like LSTM with just historical demand counts?
- Can we train a pricing or incentive model using engineered spatiotemporal features to help with surge estimation?
- Do forecast-informed repositioning assumptions lower average wait time compared to a baseline strategy in simulation?

3.2 Success Metrics

- Clustering: silhouette score, validating qualitative maps, noise rate, and runtime for inference.
- Forecasting: MAE, MSE, and MAPE on holdout time windows, stability across clusters.
- Pricing: MAE and RMSE for proxy target, feature importance sanity checks.
- Simulation: mean and median wait time reduction compared to baseline; strength under various demand patterns.

4. Work Done Till Date

4.1 Dataset and Preprocessing (Phase 1)

We used NYC taxi trip data, including pickup latitude and longitude as well as trip attributes, as a large mobility dataset. Preprocessing included limiting the data to NYC, removing invalid coordinates, and applying basic filters for trip duration and speeds. This step makes sure that clustering shows realistic pickup distributions and minimizes distortion from incorrect GPS points.

4.2 Phase 1: Hotspot Discovery and Comparative Analysis

Phase 1 tested K-Means and HDBSCAN for spatial clustering. K-Means used n_clusters = 8 and random_state = 42 to create a compact operational partition. HDBSCAN used a minimum cluster size of 30 to find density-based areas and filter out noise.

Component	K-Means	HDBSCAN
Cluster assumption	Spherical / equal variance	Arbitrary shapes; density-based
Requires K?	Yes	No (uses density hierarchy)
Noise handling	No explicit noise class	Explicit noise/outliers
Operational use	Fast inference; centroid lookup	Better offline patterns; heavier compute
Observed validity	Silhouette ~0.48	Silhouette ~0.61; ~12% noise filtered

4.2.1 Key Findings (Phase 1)

- HDBSCAN achieved better cluster validity and reduced about 12% of the spatial noise. This improved the reliability of hotspots.
- K-Means offered a quick assignment, which is good for real-time dashboard use.

- A hybrid architecture was proposed. It uses offline HDBSCAN for stable hotspot discovery and noise reduction. Online K-Means is used for fast operational mapping.

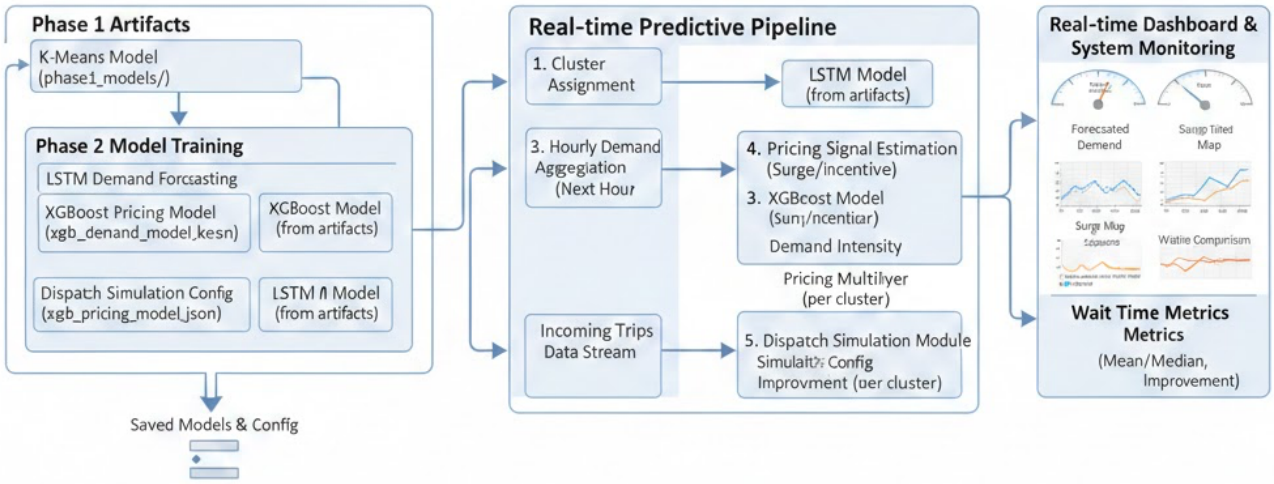
4.3 Phase 1 Dashboard (Fleet Command Center)

A Streamlit-based dashboard was created to show hotspot clusters and offer operational insights. The modules include: (i) live hotspot map visualization, (ii) research comparing clustering algorithms, and (iii) efficiency insights such as time-of-day demand heatmaps and velocity/congestion proxies. The dashboard design allows for evaluator demonstrations and future integration of predictive models in Phase 2.

4.4 Phase 2: Proposed Methodology and Mid-Term Progress

4.4.1 Overall Phase 2 System Architecture

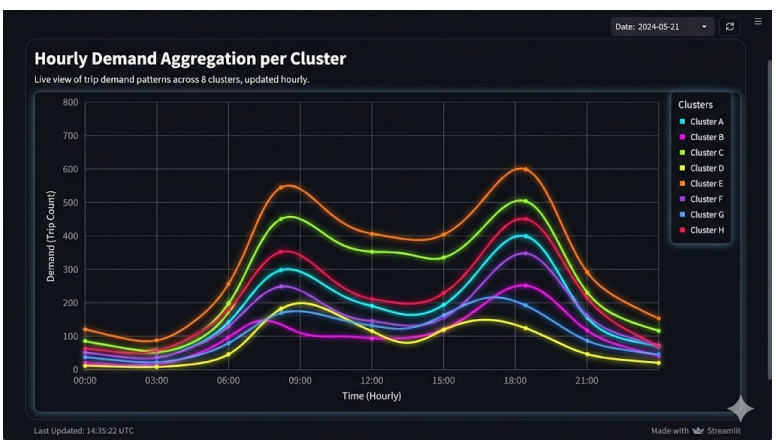
Phase 2 builds on Phase 1 artifacts, notably the saved K-Means model, to create a predictive pipeline. Incoming trips are assigned to clusters. Hourly demand per cluster is aggregated. LSTM forecasts the next hour's demand. A pricing model estimates incentive and surge signals. A simulation module evaluates the outcomes of the dispatch strategy. Models are trained offline and stored as artifacts, allowing for low-latency inference within the dashboard.

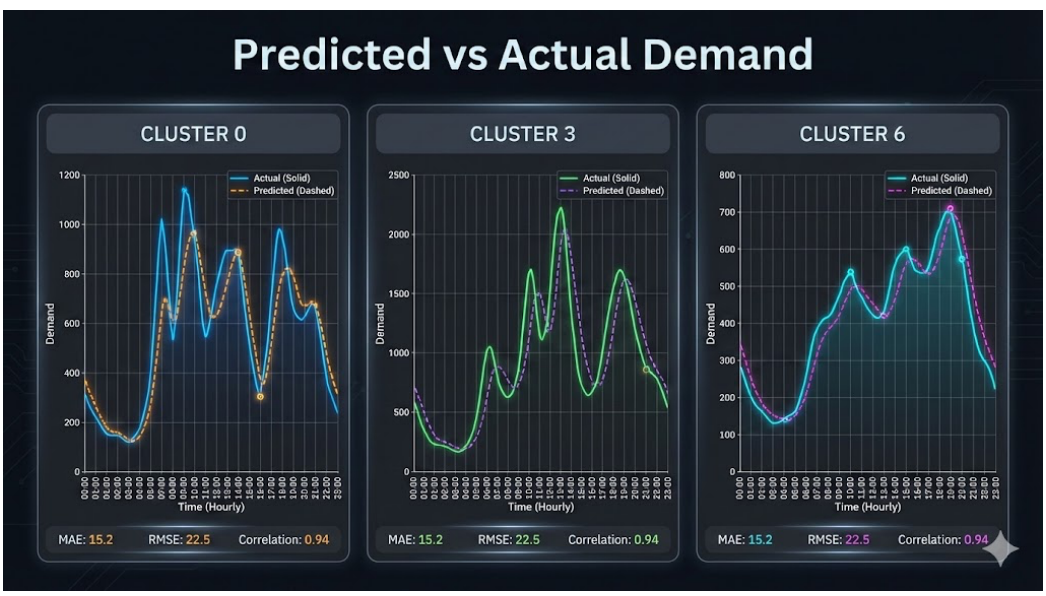


4.4.2 Demand Forecasting Module (LSTM)

Objective: Predict the demand for each cluster in the next hour to support proactive fleet rebalancing. Data pipeline: trips are assigned to clusters using the Phase 1 K-Means model. The hourly demand is calculated by counting trips per (date_hour, cluster). We create sequences using a sliding window of the last 24 hours, resulting in inputs with the shape (24, 8) and targets with the shape (8,).

- Model: LSTM (32 units, return_sequences=False), Dropout(0.2), Dense(8, ReLU)
- Optimizer: Adam. Loss: MSE. Metric: MAE.
- Training setup: 25 to 30 epochs, batch size of 32, validation split of 0.2.
- Artifacts are saved to phase2_models/ as lstm_demand_model.keras and lstm_config.json.





4.4.3 Dynamic Pricing Module (Trip Duration / Fare Proxy)

Objective: Estimate a pricing or incentive signal that matches demand intensity and operational conditions. Since direct surge labels might not be available, the mid-term approach models a proxy target like trip duration, or fare if it's available, using engineered features.

Feature Group	Example Features
Temporal	hour-of-day, day-of-week, weekend flag, cyclic sin/cos encodings
Spatial	cluster id, cluster centroid distance, pickup region density
Trip attributes	distance, passenger count (if available)
Traffic proxy	estimated speed / congestion indicator
External (optional)	weather (rain/temperature), events

Method: A Gradient Boosting or XGBoost regressor is trained offline. A surge multiplier is calculated using normalized demand intensity and the model's predicted proxy value. The resulting multiplier appears on the dashboard for each cluster and time window.

4.4.4 Dispatch Simulation Module (Discrete-Event)

Objective: Evaluate whether demand-aware positioning can reduce passenger wait time compared to baseline assumptions. The simulator creates trip requests over a day, lasting from 0 to 86400 seconds. Each request is assigned to a cluster either uniformly or based on historical or LSTM-derived demand weights. The setup includes the number of trips, drivers per cluster, average trip duration, and a random number generator seed for reproducibility.

- Baseline strategy (Random-Walk): the driver cluster is random for each request.
- AI-driven strategy: the driver is assumed to already be in the request's cluster; this is idealized hotspot positioning.

- Wait time model: wait_time = 120 + 60 × |i − j| seconds (configurable).
- Outputs: mean, median wait time for each strategy; improvement percentage.

4.5 Mid-Term Deliverables Completed

- Phase 1 pipeline and dashboard are finished. This includes a comparison of clustering and a proposal for a hybrid architecture.
- Phase 2 data pipeline defined: cluster assignment, hourly aggregation, sequence construction.
- LSTM model configuration is finalized, and the training workflow is ready. We have established a standard for saving artifacts.
- Pricing module feature design is complete. The model training plan is set using GBR and XGBoost.
- Simulation module specification is finished with measurable outputs and reproducible configuration.

5. Future Work

5.1 Experiments and Results to be Finalized

1. Run end-to-end LSTM training on the entire prepared dataset. Perform rolling-window backtesting and report MAE, MSE, and MAPE for each cluster.
2. Generate and include final plots: examples of demand aggregation, training and validation curves, predicted versus actual for multiple clusters, and error distributions.
3. Train the dynamic pricing proxy model using XGBoost or GBR. Report MAE and RMSE, and include plots showing feature importance to confirm interpretability.
4. Define surge multiplier mapping, such as piecewise linear or percentile-based approaches, and check behavior under peak and off-peak demand.

5.2 Simulation Enhancements

5. Replace the simplified cluster-distance travel time with a more realistic travel time estimate. Use centroid distance scaled by average speed or map-based ETA if APIs are available.
6. Model driver availability and service times clearly, as drivers become busy during trips and there is a limited supply per cluster.
7. Include repositioning cost due to dead mileage and report a multi-metric evaluation that combines wait time, utilization, and dead mileage.
8. Compare different policies: nearest-driver, forecast-informed repositioning, and capacity-constrained allocation.

5.3 System Integration and Final Submission

9. Integrate trained models into the Streamlit dashboard for real-time inference using stored artifacts.
10. Replace all placeholders with the final results tables and discussion. Add limitations, ethical considerations regarding pricing fairness, and future scope.

11. Prepare the final Major BTP report and presentation, including reproducible code structure, appendix, and supervisor-signed certificate.

6. References

[1] NYC Taxi & Limousine Commission (TLC), Trip Record Data (dataset/ documentation).

[2] Kaggle, “NYC Taxi Trip Duration Dataset” (dataset source).

[3] TensorFlow/Keras Documentation, LSTM, and training APIs.

[4] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” Proc. KDD, 2016.

[5] Streamlit Documentation (dashboard framework).

[6] D. Zhang et al., “A Survey on Taxi Demand Prediction,” IEEE Trans. Intelligent Transportation Systems, 2017.

[7] J. Yuan, Y. Zheng, and X. Xie, “Discovering regions of different functions in a city using human mobility and POIs,” Proc. KDD, 2012.Supervisor & Student Signatures

To be signed after final printing as required by the committee.